



The situation

An 18-footer skiff carries a large gennaker (sail) on a forward **bowsprit**. Rope tension, sail forces and a light displacement boat demand a spar that is stiff and light.



Figure 1: 18-footer with a gennaker sail attached to the bowsprit. Photo: MySailing [1].

Problem definition

Modelled as a **2D plane-stress** rectangular cantilevered beam made from aluminum with the properties $E_0 = 70$ GPa, $\nu = 0.33$, thickness $t = 0.05$ m and minimum volume fraction $V_f = 0.25$. The domain is $3.80\text{m} \times 0.50\text{m}$ ($L \times H$) with the forces: $\mathbf{F}_1 = 2$ kN at 65° on a small strip $0.175L$ to $0.225L$ and $\mathbf{F}_2 = 3$ kN at 55° on a small strip $0.95L$ to L as seen in Figure 2.

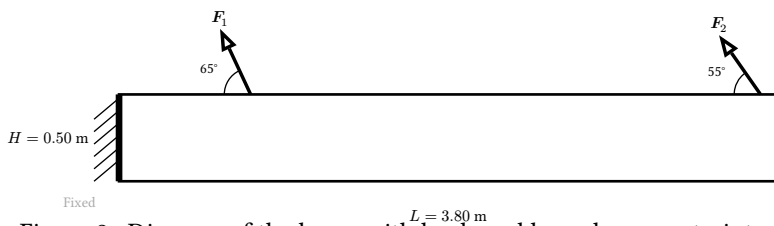


Figure 2: Diagram of the beam with loads and boundary constraints.

Elasticity BVP - weak form

The following problem is solved to get the displacement field of the bowsprit.

$$\int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) \, d\Omega = \int_{\Gamma_N} \mathbf{T} \cdot \mathbf{v} \, d\Gamma$$

Small-strain, 2D plane-stress: $\boldsymbol{\sigma} = \lambda \, \text{tr}(\boldsymbol{\varepsilon}) \mathbf{I} + 2\mu \boldsymbol{\varepsilon}$, with the Lamé parameters: $\lambda = \frac{E\nu}{1-\nu^2}$ and $\mu = \frac{E}{2(1+\nu)}$. Dirichlet BC (fixed): $\mathbf{u} = \mathbf{0}$ on Γ_D ; tractions $\mathbf{T}$ on Γ_N .

Optimization problem

The compliance is used as the cost function = external work: $J = \int_{\Gamma_N} \mathbf{T} \cdot \mathbf{u} \, d\Gamma$. The design variable $\rho \in [0, 1]$ describing density element-wise on a mesh (DG0).

$$\min_{\rho} \begin{cases} J_d + J_i + J_e & 3 \text{ realizations} \\ J_d + J_e & 2 \text{ realizations} \end{cases} \quad \text{s.t.} \quad \begin{cases} \frac{1}{LH} \int_{\Omega} \bar{\rho}_d \, d\Omega \leq V_f \\ \frac{\int_{\Omega} w \bar{\rho}_d \, d\Omega}{\int_{\Omega} w \, d\Omega} \leq V_f \\ \sigma_c^k \leq 1 \quad \forall k \in \{d, i, e\} \\ 0 \leq \rho \leq 1 \end{cases}$$

Subscripts d, i, e denote dilated, nominal, eroded realizations; $\bar{\rho}_d$ is the projected density of the dilated realization. Pitch weight $w = 1 + 2.5(\frac{x}{L})^2$ punishes mass towards the tip of the bowsprit. The constraints are evaluated on **dilated** $\bar{\rho}_d$ to guarantee all realizations satisfy limits.

SIMP - stiffness interpolation

$$E = E_{\min} + \bar{\rho}^p (E_0 - E_{\min}), \quad p = 3, \quad E_{\min} = 10^{-6} E_0$$

$p = 3$ penalises intermediate densities; $E_{\min}$ reduces numerical issues with low stiffness.

Double-filter robust formulation

By filtering once and projecting with Heaviside threshold $\eta_{\min} = \eta_d$, densities are binarized to near solid/void. A second filter is then applied and projected at different thresholds $\eta_k \in \{\eta_d, \eta_i, \eta_e\}$, producing different realizations: dilated (thicker members), nominal, or eroded (thinner members)

$$\rho \xrightarrow{\text{filter}} \tilde{\rho}_1 \xrightarrow{\eta_{\min}} \bar{\rho}_1 \xrightarrow{\text{filter}} \tilde{\rho}_2 \xrightarrow{\eta_k} \bar{\rho}_k$$

Filters solved in CG1 (radius $r = 0.04$ m) and projections in DG0 with $k \in \{d, i, e\}$.

Helmholtz PDE filter (weak form)

$$\int_{\Omega} r^2 \nabla \tilde{\rho} \cdot \nabla v \, d\Omega + \int_{\Omega} \tilde{\rho} v \, d\Omega = \int_{\Omega} \rho v \, d\Omega$$

with $\partial \tilde{\rho} / \partial n = 0$ on $\partial \Omega$. Enforces minimum feature size $\approx 2r = 0.08$ m.

Heaviside projection

$$\bar{\rho} = \frac{\tanh(\beta \eta) + \tanh(\beta(\tilde{\rho} - \eta))}{\tanh(\beta \eta) + \tanh(\beta(1 - \eta))}$$

Sharpness β ramped (1, 2, ..., 432), 15 stages, 30 MMA iters/stage: $\eta_d = 0.45$ (dilated), $\eta_i = 0.50$ (nominal), $\eta_e = 0.55$ (eroded) with $\Delta \eta = 0.05$.

Stress constraint - P-mean aggregation

Stress constraint set to $\sigma_y = 40$ MPa to account for dynamical loading:

$$\sigma_c^{p_m} = \left(\frac{1}{|\Omega|} \int_{\Omega} \left(\frac{\sigma_m}{\alpha \sigma_y} \right)^{p_m} d\Omega \right)^{\frac{1}{p_m}} \leq 1, \quad \alpha = 1$$

ε_r -relaxation removes void singularity ($\varepsilon_r = 0.2$):

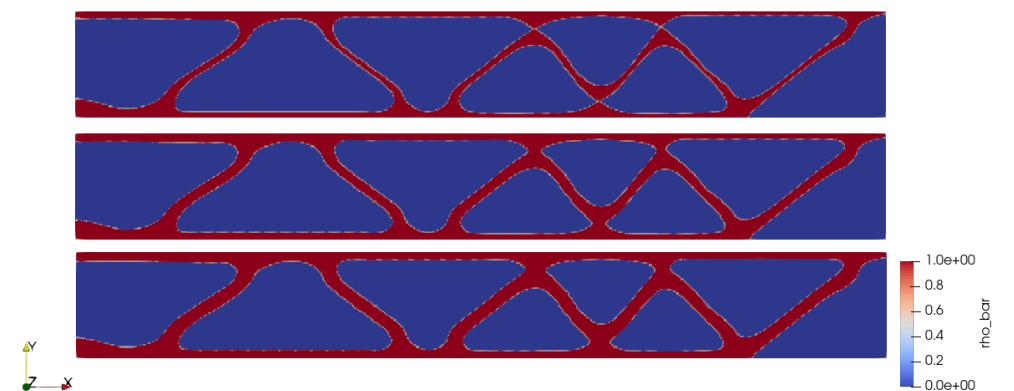
$$f_{\sigma} = \frac{\bar{\rho}}{\varepsilon_r(1 - \bar{\rho}) + \bar{\rho}}, \quad \sigma_m = f_{\sigma} \sqrt{\sigma_{vM}^2 + \sigma_{\min}^2}, \quad \sigma_{\min} = 10^{-4} \sigma_y$$

$$\sigma_{vM}^2 = \sigma_{11}^2 + \sigma_{22}^2 - \sigma_{11}\sigma_{22} + 3\sigma_{12}^2$$

β capped at $\beta \leq 2r/l_e \approx 9.6$ to prevent artificial stress concentrations. p_m -continuation: p_m ramped (2, ..., 400) over 15 stages.

Optimized designs

Optimization using 3 realizations $\{d, i, e\}$



Optimization using 2 realizations $\{d, e\}$

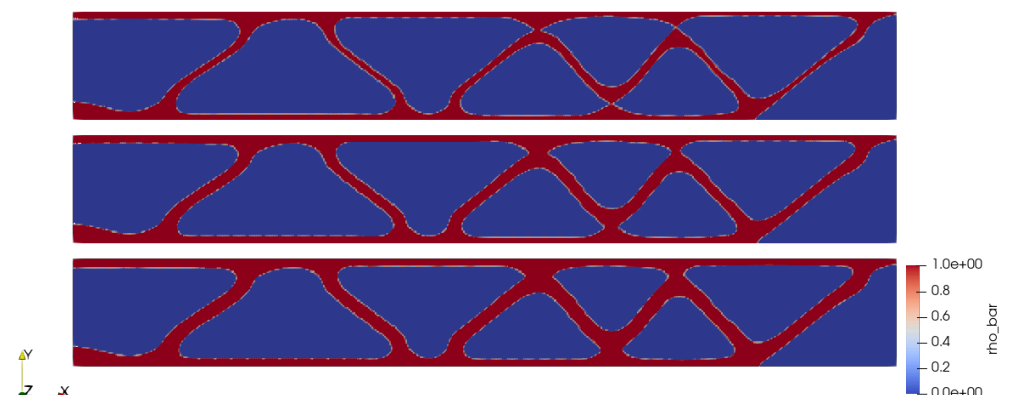


Figure 3: Optimized density fields. Top to bottom per task: eroded ($\eta_e = 0.55$), nominal ($\eta_i = 0.50$, Task 3 only), dilated ($\eta_d = 0.45$). Blue = solid, white = void.

Results & conclusion

For this optimization problem of the bowsprit, the two different approaches were almost equivalent with a negligible difference of $\approx 3.3\%$. The filter radius $r = 0.04\text{m}$ creates somewhat coarse geometries and causes some of the curved geometry at connections in the lattice structure. The effective feature size of ≈ 80 mm was chosen to ease manufacturing and reduce optimization time. The stress constraints and volume constraints were not always satisfied which a smaller filter radius could improve upon. As the stress constraints were set conservatively the geometry could still be viable.

Design	J (N·m)	V	V _p	σ_c
3R dilated	10.09	.299 [†]	.291 [†]	0.619
3R nominal	11.65	.257 [†]	.248 [†]	0.664
3R eroded	14.82	.204	.193	1.126[†]
2R dilated	10.51	.301 [†]	.295 [†]	0.654
2R interm.	12.03	.260 [†]	.253 [†]	0.729
2R eroded	15.41	.207	.199	1.152[†]

[†]constraint violated ($V_f = 0.25$, $\sigma_c \leq 1$). 3R nominal J is 3.3% lower than 2R intermediate thus including the nominal realization in optimization yields a stiffer design. Stress constraints are violated in eroded realizations and volume constraints are violated in dilated realizations.